

1. $A = x^2 - 2x + 2$, $B = 2x^2 + 2x - 2$, $C = x^2 + 3x - 2$ であるとき，次の計算をせよ。

- (1) $B + C$

答. _____
- (2) $B + 3C$

答. _____
- (3) $A + B + C$

答. _____
- (4) $-A - B + C$

答. _____
- (5) $-3A + B + C$

答. _____
- (6) $3A + B - 3C$

答. _____
- (7) $2A - B + C$

答. _____
- (8) $2A - 3B + C$

答. _____
2. 次の 1 次方程式を解け。

(1) $-2x - 5 = 8x + 8$

答. _____

(2) $-3x - 5 = x - 8$

答. _____

(3) $-3 + 2(-5x + 5) = -8x$

答. _____

(4) $9x - 4(x + 2) = 5$

答. _____

(5) $7x - 2(-x + 5) = -9$

答. _____

(6) $\frac{3}{2}x + 2 = \frac{5}{3}x + 6$

答. _____

(7) $-\frac{5}{3}x + \frac{1}{4} = -9x - \frac{5}{2}$

答. _____

(8) $8x + \frac{8}{3} = \frac{9}{4}x + 9$

答. _____

1. $A = x^2 - 2x + 2$, $B = 2x^2 + 2x - 2$, $C = x^2 + 3x - 2$ であるとき, 次の計算をせよ。

(1) $B + C$

答. $3x^2 + 5x - 4$

(2) $B + 3C$

答. $5x^2 + 11x - 8$

(3) $A + B + C$

答. $4x^2 + 3x - 2$

(4) $-A - B + C$

答. $-2x^2 + 3x - 2$

(5) $-3A + B + C$

答. $11x - 10$

(6) $3A + B - 3C$

答. $2x^2 - 13x + 10$

(7) $2A - B + C$

答. $x^2 - 3x + 4$

(8) $2A - 3B + C$

答. $-3x^2 - 7x + 8$

2. 次の 1 次方程式を解け。

(1) $-2x - 5 = 8x + 8$

$-10x = 13$

答. $x = -\frac{13}{10}$

(2) $-3x - 5 = x - 8$

$-4x = -3$

答. $x = \frac{3}{4}$

(3) $-3 + 2(-5x + 5) = -8x$

$-3 - 10x + 10 = -8x$

$-2x = -7$

答. $x = \frac{7}{2}$

(4) $9x - 4(x + 2) = 5$

$9x - 4x - 8 = 5$

$5x = 13$

答. $x = \frac{13}{5}$

(5) $7x - 2(-x + 5) = -9$

$7x + 2x - 10 = -9$

$9x = 1$

答. $x = \frac{1}{9}$

(6) $\frac{3}{2}x + 2 = \frac{5}{3}x + 6$

両辺に 6 をかけて, $9x + 12 = 10x + 36$

答. $x = -24$

(7) $-\frac{5}{3}x + \frac{1}{4} = -9x - \frac{5}{2}$

両辺に 12 をかけて, $-20x + 3 = -108x - 30$

答. $x = -\frac{3}{8}$

(8) $8x + \frac{8}{3} = \frac{9}{4}x + 9$

両辺に 12 をかけて, $96x + 32 = 27x + 108$

答. $x = \frac{76}{69}$